

Radiotherapy for cervical cancer

You may have internal radiotherapy (brachytherapy) or external radiotherapy to treat cervical cancer.

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About radiotherapy for cervical cancer

[Radiotherapy](#) treats cancer by using high-energy x-rays. These destroy the cancer cells while doing as little harm as possible to normal cells.

You may have radiotherapy:

- if you have [early or locally advanced](#) cervical cancer
- after [surgery](#), if there is a high risk of the cancer coming back
- if cancer comes back in the pelvis after surgery
- to help relieve symptoms such as bleeding.

Radiotherapy for [cervical cancer](#) may be given:

- externally, from a machine outside the body
- internally, from radioactive material that is temporarily put into the part of the body being treated – doctors call this brachytherapy.

You may have both external and internal radiotherapy. The doctor who plans your treatment will discuss this with you.

Chemoradiation

Radiotherapy for cervical cancer is often given with chemotherapy. This is called [chemoradiation](#). The chemotherapy drugs make the cancer cells more sensitive to radiotherapy. The combination of treatments can be more effective than having radiotherapy on its own.

The chemotherapy drug most commonly used is [cisplatin](#). It is usually given once a week throughout your radiotherapy.

The side effects of chemoradiation are similar to [radiotherapy side effects](#). But they can be more severe. Your doctor, radiographer or specialist nurse can give you more information about chemoradiation and the possible side effects of treatment.

Early menopause

Radiotherapy for cervical cancer affects the ovaries. If you are still having periods, radiotherapy will cause an early [menopause](#). You may have a period during the course of radiotherapy, but your periods will stop after treatment finishes. Your healthcare team will discuss this with you before your treatment starts. They can also give you information about treatments to [manage menopausal symptoms](#).

Ovarian transposition

It is possible to have an operation to move the ovaries. The ovaries are lifted out of the radiation field and attached to the abdomen (tummy) wall. Doctors call this ovarian transposition. You have it before radiotherapy starts. The aim is to prevent an early menopause.

Having ovarian transposition depends on the stage of the cancer and the risk that it has spread to the ovaries. Your cancer doctor can talk to you about this.

Ovarian transposition is usually done using [laparoscopic \(keyhole\) surgery](#). Sometimes the ovaries are moved at the same time as a [hysterectomy](#) for cervical cancer. The surgeon may do this if they think you might need radiotherapy after surgery. Ovarian transposition does not always work, and cervical cancer treatment can still cause early menopause.

Fertility

Radiotherapy for cervical cancer also affects the womb. After treatment with radiotherapy, the womb is not able to carry a baby. If you would like to get pregnant in the future, your cancer doctor can refer you to a fertility specialist. They can see you before you begin treatment to talk through your [fertility options](#).

External radiotherapy

External beam radiotherapy is given from a radiotherapy machine outside the body.

We have more information about:

- [how radiotherapy is planned](#)
- [how external beam radiotherapy is given](#).

For cervical cancer, you usually have this treatment as an outpatient, once a day from Monday to Friday with a rest at the weekend. Each session of treatment takes a few minutes. It usually takes about 5 to 5 ½ weeks to have the full course of treatment.

Cervical cancer is often treated with a type of external beam radiotherapy called image-guided radiotherapy (IGRT). This means that, as well as a first planning visit, you have further assessments done at each treatment session. Images are taken before each treatment to check the size and position of the cancer. If there are any changes, adjustments can be made to the treatment.

Internal radiotherapy (brachytherapy)

Internal radiotherapy is called brachytherapy. It gives radiation directly to the cervix and the area close by. It is usually given after external radiotherapy.

You have one or more hollow tubes, called applicators, put into your womb or vagina. The radiotherapy is given through these tubes. How you have your treatment depends on whether you have had your womb removed.

In this video, Sam, a Therapeutic Radiographer, explains what brachytherapy is and how it is used to treat cervical cancer, while Nicky shares her experience of treatment and the support she received from her radiotherapy team.



Brachytherapy if you have not had a hysterectomy

If you have not had a hysterectomy, you will have intrauterine brachytherapy (radiotherapy given into the womb). You have a general anaesthetic or a spinal anaesthetic first. Your doctor will explain this to you.

Your doctor inserts applicators into your vagina. They pass them up through the cervix into the womb. They may also place applicators alongside the cervix.

The doctor may place padding inside the vagina. This is to help protect your back passage (rectum) and prevent the applicators moving. You also have a catheter put into your bladder to drain urine.

The applicators can be uncomfortable, so you may need to take painkillers while they are in.

Image: Cervical brachytherapy

Brachytherapy if you have had a hysterectomy

If you have had a hysterectomy, a doctor may place one or two applicators into the vagina. This is called vaginal brachytherapy.

Vaginal brachytherapy is a simpler treatment than brachytherapy given into the womb. You do not need any preparation for the treatment. It is unlikely to cause any immediate side effects.

Having brachytherapy treatment

You have a scan to check the position of the applicators. When it is confirmed that the applicators are in the right position, they are connected to the brachytherapy machine. The machine is operated by a radiographer. It places a radioactive capsule, called a source, into the applicators. The machine then gives the planned dose of radiation.

Internal radiotherapy can be given as high-dose-rate, low-dose-rate or pulsed-dose-rate treatment. Most centres in the UK use high-dose-rate equipment. These different ways of giving internal radiotherapy all work well. The type you have will depend on the system your hospital uses. Your cancer specialist and specialist nurse will explain more so that you know what to expect.

High-dose-rate treatment



This is the most common way of giving brachytherapy to the cervix. You have several treatments. Each treatment is given over about 10 to 15 minutes. How high-dose-rate treatments are given varies from hospital to hospital.

If you stay in hospital, you have your treatments over several days. The applicators may be removed between treatments. Or they may be left in place and removed after your final treatment.

If you have your treatment as an outpatient, you go to the hospital 3 or 4 times over several days or a week. A nurse will remove the applicators before you go home.

You may have a tube (catheter) put into your bladder to drain urine during high-dose rate treatment. A nurse will take this out before you go home.

Low-dose treatment



If you have this treatment, you usually stay in hospital for 12 to 24 hours. But sometimes it is given over a few days. Your doctor, nurse or radiographer will tell you more about low-dose-rate treatment.

Pulsed-dose-rate brachytherapy



This treatment is given over the same length of time as low-dose-rate treatment. But the radiation dose is given in pulses, rather than as a continuous dose. Your doctor, nurse or radiographer will give you more information.

In this video, Sam, a Therapeutic Radiographer, explains the removal of the brachytherapy applicator after cervical cancer treatment.

 [Removing the brachytherapy applicator | cervical cancer](#)

Side effects of radiotherapy

You may develop side effects during your treatment. We have more information about possible [pelvic radiotherapy side effects](#). These usually improve over a few weeks or months after treatment finishes. Your doctor, nurse or radiographer will discuss this with you, so you know what to expect. Tell them about any side effects you have during or after treatment. There are often things that can help.

Smoking makes the side effects of radiotherapy worse. If you smoke, stopping smoking will help. If you want help or advice on how to give up, talk to your clinical oncologist, GP or a specialist nurse. The NHS has information and support to help you [give up smoking](#).

We have more information about [late effects of pelvic radiotherapy](#).

Support from Macmillan

Macmillan is here to support you. If you would like to talk, you can do the following:

- Call the Macmillan Support Line for free on [0808 808 00 00](#).
- [Chat to our specialists online](#).
- Visit our [radiotherapy and side effects forum](#) to talk with people who have been affected by radiotherapy, share your experience, and ask an expert your questions.

About our information

References



Below is a sample of the sources used in our cervical cancer, cervical screening and CIN information. If you would like more information about the sources we use, please contact us at canцерinformationteam@macmillan.org.uk

GOV.UK. Cervical screening: programme overview. Updated 18 November 2019. Available from www.gov.uk/guidance/cervical-screening-programme-overview (accessed March 2020).

GOV.UK. Colposcopic diagnosis, treatment and follow up. Updated 5 February 2020. Available from www.gov.uk/government/publications/cervical-screening-programme-and-colposcopy-management/3-colposcopic-diagnosis-treatment-and-follow-up (accessed April 2020).

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Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by Senior Medical Editor, [Professor Nick Reed](#), Consultant Clinical Oncologist.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English

- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

Date reviewed

Reviewed: 01 April 2021 | Next review: 01 April 2024

ⓘ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help